# SV16 to-do list - rebuild step 1

Use the step-1 copy SBSV-16_rebuild_step1_values.kicad_pcb as the starting file. Tick each item when done. Items marked (not bought) need purchase before the board can work.

## A. Footprint changes in KiCad (Footprint Editor / Change Footprint)

[ ] C25, C30: change footprint from C_0805 (ceramic) to CP_Elec_4x5.4 (electrolytic), value 10 uF 35 V.

[ ] C32: change footprint from C_0805 to C_0603 (470 nF, spec: 0603).

[ ] R9, R10, R11, R12 (100 ohm): change footprint from R_1206 to R_0805 (bought 100 ohm pack is 0805).

[ ] L1, L2 (10 uH CD54): footprint is L_Sunlord_SWPA6045S. Spec says L_Sunlord_MWSA0518 (5.4x5.2 mm). Confirm the footprint against the bought part datasheet and update.

[ ] J7 (2x25 spare header): add footprint Connector_PinHeader_2.54mm:PinHeader_2x25_P2.54mm_Vertical, mark DNP / not fitted.

[ ] D6 (red LED, INITN): mark DNP (spec: do not fit by default).

[ ] FB1 (600 ohm ferrite): change footprint from 0603 to 0805 (Inductor_SMD:L_0805_2012Metric). Part in cart: TDK MPZ2012S601ATD25, 0805, qty 2.

[ ] R21, R22, R23, R24 (0 ohm): no 0 ohm bought. Use solder-bridge pads. Spec: populate one pair only (R21/R23 OR R22/R24).

[ ] R5 and R25 (4.7k): change footprint from R_0603 to R_0805, and use the 4.7k 0805 pack you bought.

[ ] R6, R7, R8 (1k): change footprint from R_0603 to R_1206, and use the 1k 1206 pack you bought.

[ ] Y2 (12 MHz): change footprint from Crystal_HC49-4H (through-hole) to the SMD HC49/US footprint for the bought 12 MHz surface-mount crystal (2-pin, passive, OK for CH340G).

[ ] C36, C37 (22 pF): change footprint from C_0603 to C_0805, to match the YAGEO CC0805JRNPO9BN220 (22 pF C0G 0805 50 V 5%) you are buying.

[ ] R47 (10k): change footprint from R_1206 to R_0603. Use the 5% 0603 part (YAGEO RC0603JR-0710KL), already in cart.

## B. Value decisions still open (you decide, then set the value in KiCad)

[ ] R44: no value in spec or board. Leave unset and report it.

[ ] R45: 100k on board, not in spec. Probably unnecessary. Remove unless the schematic shows a use.

[ ] R48, R49: AP62300 EN divider from VM_IN. Keep 100k (R48) and 33k (R49). At 12 V, EN = 2.98 V, under the 6 V EN pin limit. AP62300 EN threshold is 1.2 V typ (1.25 V max) rising, so the buck starts at about 4.7 to 4.9 V VIN (minimum VIN is 4.2 V). Checked against the Diodes DS41958 datasheet.

[ ] C34: 100 nF on board, not in spec. Probably unnecessary. Remove unless the schematic shows a use.

[ ] C35: 100 pF on board, not in spec. Probably unnecessary. Remove unless the schematic shows a use.

## C. Values already changed in the step-1 copy (verify in KiCad)

[ ] Q2: BC817 changed to MMBT3904. Q4: BC807 changed to MMBT3906 (SOT-23, same pinout per spec).

[ ] C36, C37: 100 nF changed to 22 pF (CH340G crystal load caps, spec).

[ ] Resistors: values set per spec (see BOM). R1, R13-R19, R26, R32, R47 = 10k; R2-R5, R25 = 4.7k; R6-R8, R31, R33, R34 = 1k; R9-R12 = 100 ohm; R27-R30 = 470 ohm; R36, R37 = 5.1k; R38 = 12.4k; R39, R46 = 33k; R40, R41, R43 = 100k.

[ ] Check the file opens with all 141 footprints and that the tracks are unchanged (this copy was checked: 141 footprints, 1,783 segments, 179 vias).

## D. Schematic / design checks from the spec (must be fixed in the schematic)

[ ] U5 (AP62300) pin 4 (FB) must be connected to 3V3. Spec says the earlier sheet left it floating.

[ ] U6 (MP1584) COMP: R3 = 150 kohm in series with C3 = 100 pF to GND (COMP pin 3). Calculated with the MP1584EN datasheet method: fsw 900 kHz (R40 = 100k), fc 90 kHz, C2 = 100 uF, Vout 1.1 V. Check: zero at 10.6 kHz, below fc/4. C6 (second cap, COMP to GND) is needed if the output ESR zero is below fsw/2, which is likely with the tantalum C21. C6 value is not calculated: confirm the output ESR and the datasheet C6 equation before fitting.

[ ] U6 feedback divider: R38 = 12.4k top, R39 = 33k bottom (already set). Keep R39 below 40k.

[ ] C23: change to 470 uF 25 V electrolytic (same CP_Elec_8x10.5 footprint). 16 V on 12 V is 75% of rated, too close to the limit.

[ ] Oscillator Y1 (ECS-3225MV-250): pin 1 Tri-state (high or NC = active), pin 2 GND, pin 3 OUT, pin 4 VDD. Confirm the pad footprint against the ECS land pattern.

[ ] Y2 (12 MHz): check the footprint against the bought crystal (HC49-4H vs 3225 SMD).

[ ] LP5907 output cap (C42): electrolytic is used here, spec wants low-ESR ceramic. Check stability on the bench.

## E. Purchases still needed (not in any cart screenshot)

[ ] U1: FPGA LFE5U-12F-6TG144C (not bought).

[ ] U3: 1 more W25Q64JVSSIQ (you have 1, need 2).

[ ] Y1: ECS-3225MV-250, active 25 MHz XO 3.2x2.5 4-pad. The 3 bought 2-pin 25 MHz crystals cannot be used here.

[ ] Y2: 12 MHz SMD crystal HC49/U (surface mount, 2-pin). To buy. Footprint change needed (see A).

[ ] D8/D9/D11: 1 more SS34 (you have 2, need 3).

[ ] 10k 0603 5% resistors: YAGEO RC0603JR-0710KL, 11 pieces (R1, R13-R19, R26, R32). In cart (qty 110, covers R47 too).

[ ] 10k 0603 5% resistor: YAGEO RC0603JR-0710KL, 1 piece for R47 (12 in total with the 11 above). Covered by the cart line. The user uses 5% parts only.

[ ] 22 pF 0805 C0G caps: YAGEO CC0805JRNPO9BN220 (22 pF, 50 V, 0805, 5%), 2 pieces (C36, C37). To buy.

[ ] 4.7k (R5, R25) and 1k (R6-R8): use the 0805 and 1206 packs you already bought, with the footprint changes in section A. No purchase needed.

[ ] Headers: use the most available part, standard 2.54 mm straight male pin headers (1x10, 2x10, 1x6, 1x4, 1x2). Footprints on the board already match, so no change. J2 (1.27 mm 2x5) left unpopulated per spec. J6 is the exception: use a shrouded, keyed 1x6 header (not a standard one).

[ ] Connectors and switches (barrel jack J9, SW1/SW2, JP1 shunt): you already have these. Not in the cart screenshots, so confirm the parts and footprints match the board.

## G. Agent handoff (for whoever makes the PCB changes)

[ ] Starting file: SBSV-16_rebuild_step1_values.kicad_pcb (in this folder). It has the same 141 footprints, positions and 4-layer stackup as SBSV-16.kicad_pcb, with 37 values changed. Do not move components. Do not change pad numbers or net names without the decision in G3.

[ ] G1 Exact KiCad footprint names (verify each exists in your library before use): Y1 current Oscillator:Oscillator_SMD_Abracon_ASE-4Pin_3.2x2.5mm. Bought part is ECS-3225MV-250, a 4-pad 3.2x2.5 mm XO.

Board pads are 1.3 x 1.1 mm at (+-1.05, +-0.825). The ECS suggested land pattern labels read as pads 1.4 x 1.2 mm at (+-1.10, +-0.80) (read from figure labels, verify against the ECS drawing). Replace the footprint with the ECS pattern before routing. Y2 change to Crystal:Crystal_SMD_HC49-SD (or the footprint for the bought SMD HC49/U crystal). L1, L2 change to Inductor_SMD:L_Sunlord_MWSA0518_5.4x5.2mm (name from spec; confirm pads against the bought CD54 datasheet). C25, C30 Capacitor_SMD:CP_Elec_4x5.4. C32 Capacitor_SMD:C_0603_1608Metric. C36, C37 Capacitor_SMD:C_0805_2012Metric. R5, R25 Resistor_SMD:R_0805_2012Metric. R6, R7, R8 Resistor_SMD:R_1206_3216Metric. R9-R12 Resistor_SMD:R_0805_2012Metric. R47 Resistor_SMD:R_0603_1608Metric. R21-R24 solder-bridge pads (or R_0603 footprint with 0 ohm). J7 Connector_PinHeader_2.54mm:PinHeader_2x25_P2.54mm_Vertical, not fitted. D6 mark DNP.

[ ] G2 Y1 pinout (ECS-3225MV datasheet): pin 1 Tri-state (high or NC = active, low = high-Z), pin 2 GND, pin 3 OUT, pin 4 VDD. Map pads 1-4 to these before routing.

[ ] G3 Net-name decisions (choose one per row before routing): USB pair: board uses USB_DP/USB_DN, spec uses USB_D+/USB_D-. Decision: file names kept, spec names recorded in REPORT.md. 5 V rail: board uses 5V_USB, spec uses 5V (J3/J4 pins 17/18). Decision: file name 5V_USB kept, spec name 5V recorded in REPORT.md. All other spec nets (clk_25m, ext_rst_n, uart_rx/tx, flash_*, spi0_*, VM_IN, 3V3, 2V5, 1V1, GND) are already in the file.

[ ] G4 Design rules (spec section 7): signal track 0.25 mm; 3V3, 1V1, 2V5 power tracks 0.8 mm; VM_IN 1.5 mm; switching nodes (U5 OUT-L1, U6 SW-L2) 0.8 mm and short. Clearance 0.2 mm minimum, 0.5 mm between switching nets and JTAG/flash nets. Vias 0.2 mm drill / 0.4 mm pad; power vias 0.3 / 0.6 mm. USB_D+/USB_D- 90 ohm differential, 0.2 mm gap, length matched within 0.5 mm. FPGA decoupling caps C2-C20 within 3 mm of their pin, via first at the pad. Each FPGA GND pin gets its own via to the GND plane. Y1 within 10 mm of pin 133, no via in the clock trace, ground pour under it, 5 mm or more from switching nodes. U2 within 15 mm of FPGA pins 46/47/49/54; R9-R12 at the FPGA end. J6 GND return separate from FPGA ground. No copper pour under USB-C shield pads or barrel jack body. TP1-TP6 on top side.

[ ] G5 Strip the old routing first: the uploaded file still has 1,783 tracks and 179 vias from the old routing. Delete all tracks and vias in KiCad (select all, filter to tracks and vias, delete). Keep every footprint in place. Do not merge or copy old routing (routing_state.json, routing.kicad_pcb.txt, sv16_board_working.kicad_pcb). Confirm the board then shows 0 tracks and 0 vias and all nets unconnected.

[ ] G6 Acceptance: after routing, DRC shows 0 unconnected, 0 shorts, 0 clearance, 0 edge and courtyard errors. Zones filled. Only then generate Gerber, drill, BOM and position files.

## F. Routing and checks (later step, not part of this netted board)

[ ] Re-route the board from the updated placement. Tracks in this file are not verified.

[ ] Run DRC: 0 unconnected, 0 shorts, 0 clearance, 0 edge or courtyard errors. Zones filled.

[ ] Only after clean DRC: generate Gerber, drill, BOM and position files.

[ ] Do not send the board to fabrication until all of the above are done.